Phase 1 Project

This project is designed to **analyze a comprehensive flight dataset** that includes:

```
    - accident dates,
    - aircraft types,
    - registrations,
    - operators,
    - fatalities,
    - locations,
    - and damages,
```

By examining patterns in fatal and non-fatal accidents, damage severity, and aircraft types, the project aims to **help investors select aircraft with the optimal balance of safety and financial risk**, enabling data-driven decisions that minimize exposure to catastrophic losses while maximizing long-term return on investment.

Data - from kaggle (https://www.kaggle.com/datasets/anandkushawaha/aviation-crashed-flights-data?resource=download)

1. Setup & Configuration

```
# Import important librabries needed

# Data Handling
import pandas as pd
import numpy as np

# Visualization
import matplotlib
import matplotlib.pyplot as plt

# Configure notebook display settings
%matplotlib inline
plt.rcParams['figure.figsize'] = (12, 8) # Set default figure size for
plots
```

2. Data Ingestion

```
# Path to the dataset obtained from Kaggle

# Path to Excel file
file_path = './flight.csv'

# Load the dataset into a pandas DataFrame
flight_df = pd.read_csv(file_path, index_col=0) # remove index column
if not needed
```

```
flight_df.head()  # Display the first few rows of the DataFrame
```

```
      acc.date                                    type       reg  \
0   3 Jan 2022  British Aerospace 4121 Jetstream 41    ZS-NRJ
1   4 Jan 2022  British Aerospace 3101 Jetstream 31    HR-AYY
2   5 Jan 2022                       Boeing 737-4H6    EP-CAP
3   8 Jan 2022                 Tupolev Tu-204-100C  RA-64032
4  12 Jan 2022        Beechcraft 200 Super King Air       NaN

                                      operator fat  \
0                                   SA Airlink   0
1  LANHSA - Línea Aérea Nacional de Honduras S.A   0
2                              Caspian Airlines   0
3                        Cainiao, opb Aviastar-TU   0
4                                       private   0

                                          location  dmg
0                          near Venetia Mine Airport  sub
1  Roatán-Juan Manuel Gálvez International Airpor...  sub
2              Isfahan-Shahid Beheshti Airport (IFN)  sub
3      Hangzhou Xiaoshan International Airport (HGH)  w/o
4     Machakilha, Toledo District, Grahem Creek area  w/o
```

3. Data Profiling / Initial Inspection

```
# Print the shape of the DataFrame (rows, columns)
print(f"df shape: {flight_df.shape}")
```

```
df shape: (2500, 7)
```

3.1 Check for null values and fix it

```
flight_df.isnull().sum()
```

```
acc.date     0
type         0
reg         92
operator    14
fat         12
location     0
dmg          0
dtype: int64
```

We have a 92 missing values from reg, 14 from opertor and 12 from fat. We have two options:

1. Fill in missing values:
   - Relpace the missing values with appropriate "Unkown" that enables us to maintain data integrity and allows us to keep all rows without loosing data for better analysis

2. Drop Rows with missing values
   - Remove all rows with missing data that keeps the analysis clean and focused on known data. The downside is that I lose potentially valuable data and reduce the dataset size

In this project, **option 2** is the best due to reliability.

3.1.1 Working on missing Values for Reg

```
# Sample 5 rows where 'reg' column is not null
has_reg = flight_df[flight_df["reg"].notna()].sample(5)
has_reg
```

```
        acc.date                                    type     reg
operator  \
17   14 Feb 2022                    Let L-410UVP-E3  9S-GFA  Doren
Air Congo
958  22 Dec 2019                 Boeing 737-824 (WL)  N87513  United
Airlines
191  17 Nov 2022                    Beechcraft 1900D  VH-NYA   Penjet
Pty Ltd
534   8 May 2020                    Beechcraft 1900C  N31704
Ameriflight
386  13 Oct 2021  Bombardier BD-700-1A10 Global 6000  9H-VJM
VistaJet

    fat                                          location  dmg
17    0                        Bukavu-Kavumu Airport (BKY)  w/o
958   0        Denver International Airport, CO (DEN/KDEN)  sub
191   0                Fortnum Airport (YFOR), Fortnum, WA  sub
534   0        San Antonio International Airport, TX (SAT)  sub
386   0  Johannesburg-O.R. Tambo International Airport ...  sub
```

```
# Sample 5 rows where 'reg' column is null
no_reg = flight_df[flight_df["reg"].isna()].sample(5)
no_reg
```

```
        acc.date                             type  reg operator fat  \
928  23 Nov 2019                  Boeing 737-800  NaN      NaN   0
245  21 Feb 2021   Beechcraft 200 Super King Air  NaN  private   0
806  13 Jul 2019  Beechcraft B200 Super King Air  NaN  Unknown   0
777  13 Jun 2019  Beechcraft B200 Super King Air  NaN  Unknown   0
609  18 Sep 2020                         Learjet  NaN  private   0

                        location  dmg
928  Nürnberg Airport (NUE/EDDN)  non
245           San Andrés, Petén  w/o
806        Graham Creek, Toledo  w/o
777      Río Chixoy, La Máquina  w/o
609                      Zanja  w/o
```

```
# Drop rows with missing 'reg' values
flight_df = flight_df.dropna(subset=['reg'])

# Check for remaining missing values
flight_df["reg"].isna().sum()
```

```
np.int64(0)
```

```
# Assertain that there are no missing values in 'reg' column
assert flight_df["reg"].isna().sum() == 0
```

3.1.2 Working on missing Values for Operator

```
# Sample 5 rows where 'operator' column is  not null
has_operator = flight_df[flight_df["operator"].notna()].sample(5)
has_operator
```

```
         acc.date                                type     reg
operator  \
189   15 Nov 2022   Fairchild SA227-AT Expediter  N247DH
Ameriflight
573   17 Jul 2020  Beechcraft 200 Super King Air  C-FSKQ     Skyjet
Aviation
852   18 Aug 2019            Boeing 757-251 (WL)  N543US     Delta Air
Lines
103    1 Jul 2022            Boeing 737-7H4 (WL)  N480WN  Southwest
Airlines
1034   4 Apr 2018                  Learjet 45XR  N618CW    ERG
Holdings LLC

     fat                                              location  dmg
189    0                                          Pewaukee, WI  sub
573    0                              Rouyn Airport, QC (YUY)  sub
852    0  Ponta Delgada-João Paulo II Airport, Azores (PDL)  sub
103    0  Santa Ana-John Wayne International Airport, CA...  non
1034   0         Houston-William P. Hobby Airport, TX (HOU)  w/o
```

```
# Sample rows where 'operator' column is null
no_operator = flight_df[flight_df["operator"].isna()]
no_operator
```

```
         acc.date             type     reg operator fat
location  dmg
1015  22 Mar 2018  Airbus A320-232  VH-...      NaN   0  near Cairns,
QLD  non
1015  22 Mar 2018  Airbus A320-232  VH-...      NaN   0  near Cairns,
QLD  non
```

```
# Drop rows with missing 'operator' values
flight_df = flight_df.dropna(subset=['operator'])
```

```
# Check for remaining missing values
flight_df["operator"].isna().sum()
```

```
np.int64(0)
```

```
# Assertain that there are no missing values in 'operator' column
assert flight_df["operator"].isna().sum() == 0
```

3.1.3 Working on missing Values for Fat

```
# Sample 5 rows where 'fat' column is not null
has_fat = flight_df[flight_df["fat"].notna()].sample(5)
has_fat
```

```
         acc.date                       type     reg  \
221    9 Jan 2021    Cessna 560 Citation V    N3RB
1090  10 Jun 2018         Boeing 777-223ER  N750AN
317    4 Jul 2021          Airbus A321-231  N156AN
1092  10 Jun 2018  Boeing 787-9 Dreamliner  N826AN
346   12 Aug 2021     Shijiazhuang Y-5B(D)  B-8440

                            operator fat
location  \
221                  SX Transport LLC   1           23 km SE of Pine
Grove, OR
1090               American Airlines   0
Decatur, Texas
317                American Airlines   0                        Miami,
Florida
1092               American Airlines   0                           near
Harbin
346   Northeast General Aviation Co.   0  Nengbei Farm, Heilongjiang
Province

      dmg
221   w/o
1090  non
317   non
1092  non
346   w/o
```

```
# Sample 5 rows where 'fat' column is null
no_fat = flight_df[flight_df["fat"].isna()].sample(5)
no_fat
```

```
         acc.date                                type       reg  \
658    5 Dec 2020  British Aerospace BAe-125-800A    N484AR
30    24 Feb 2022                   Antonov An-26  RF-36074
30    24 Feb 2022                   Antonov An-26  RF-36074
1223  23 Nov 2018  British Aerospace BAe-125-700A     N422X
```

```
521    5 Apr 2020                    Antonov An-26  UP-AN601

                  operator  fat                             location
dmg
658                private  NaN                Jesús María Semprúm
w/o
30       Russian Air Force  NaN  near Ostrogozhsk, Voronezh Region
w/o
30       Russian Air Force  NaN  near Ostrogozhsk, Voronezh Region
w/o
1223               private  NaN       S of Curaçao [Caribbean Sea]
mis
521   Libyan National Army  NaN                         near Tarhuna
w/o
```

```
# Drop rows with missing 'fat' values
flight_df = flight_df.dropna(subset=['fat'])

# Check for remaining missing values
flight_df["fat"].isna().sum()
```

```
np.int64(0)
```

```
# Assertain that there are no missing values in 'operator' column
assert flight_df["fat"].isna().sum() == 0
```

Confirm that there are no more null values

```
flight_df.isnull().sum()
```

```
acc.date    0
type        0
reg         0
operator    0
fat         0
location    0
dmg         0
dtype: int64
```

4. Identify Text Data that Require Cleaning

```
flight_df["type"].value_counts()
```

```
type
Cessna 208B Grand Caravan                   108
Antonov An-2R                                58
Beechcraft 200 Super King Air                40
de Havilland Canada DHC-6 Twin Otter 300     34
de Havilland Canada DHC-8-402Q Dash 8        28
                                           ...
```

```
Boeing 737-86J (WL)                              2
Boeing 767-375ER                                 2
Boeing 747-412 (BCF)                             2
Boeing 747-412F (SCD)                            2
Rockwell Sabreliner 80                           2
Name: count, Length: 506, dtype: int64
```

There are cases where we have data entry issues, and publishers that should be encoded the same have not been. The specification of Boeing 737-86J(WL) AND bOEING 767-375ER, for example can be made into the same categories

```
# Remove leading/trailing whitespace from 'type' column
# Replace all variations to unify category names
flight_df['type'] = (
    flight_df['type']
    .str.strip()
    .replace(r'^Airbus.*', 'Airbus', regex=True)
    .replace(r'^Antonov.*', 'Antonov', regex=True)
    .replace(r'^ATR.*', 'ATR', regex=True)
    .replace(r'^Basler.*', 'Basler', regex=True)
    .replace(r'^Beechcraft.*', 'Beechcraft', regex=True)
    .replace(r'^Boeing.*', 'Boeing', regex=True)
    .replace(r'^Bombardier.*', 'Bombardier', regex=True)
    .replace(r'^British.*', 'British Aerospace', regex=True)
    .replace(r'^Britten.*', 'Britten-Norman', regex=True)
    .replace(r'^Canadair.*', 'Canadair', regex=True)
    .replace(r'^CASA.*', 'CASA', regex=True)
    .replace(r'^Cessna.*', 'Cessna', regex=True)
    .replace(r'^Cirrus.*', 'Cirrus', regex=True)
    .replace(r'^Convair.*', 'Convair', regex=True)
    .replace(r'^Dassault.*', 'Dassault', regex=True)
    .replace(r'^De Havilland.*', 'De Havilland', regex=True)
    .replace(r'^Dornier.*', 'Dornier', regex=True)
    .replace(r'^Douglas.*', 'Douglas', regex=True)
    .replace(r'^Eclipse.*', 'Eclipse', regex=True)
    .replace(r'^Embraer.*', 'Embraer', regex=True)
    .replace(r'^Fairchild.*', 'Fairchild', regex=True)
    .replace(r'^Fokker.*', 'Fokker', regex=True)
    .replace(r'^Grumman.*', 'Grumman', regex=True)
    .replace(r'^Gulfstream.*', 'Gulfstream', regex=True)
    .replace(r'^Harbin.*', 'Harbin', regex=True)
    .replace(r'^Hawker.*', 'Hawker', regex=True)
    .replace(r'^IAI.*', 'IAI', regex=True)
    .replace(r'^Ilyushin.*', 'Ilyushin', regex=True)
    .replace(r'^Learjet.*', 'Learjet', regex=True)
    .replace(r'^Let.*', 'Let', regex=True)
    .replace(r'^Lockheed.*', 'Lockheed', regex=True)
    .replace(r'^McDonnell.*', 'McDonnell Douglas', regex=True)
    .replace(r'^Pilatus.*', 'Pilatus', regex=True)
```

```
    .replace(r'^Raytheon.*', 'Raytheon', regex=True)
    .replace(r'^Rockwell.*', 'Rockwell', regex=True)
    .replace(r'^Saab.*', 'Saab', regex=True)
    .replace(r'^Shaanxi.*', 'Shaanxi', regex=True)
    .replace(r'^Shijiazhuang.*', 'Shijiazhuang', regex=True)
    .replace(r'^Shorts.*', 'Shorts', regex=True)
    .replace(r'^Swearingen.*', 'Swearingen', regex=True)
    .replace(r'^Viking.*', 'Viking', regex=True)

    )

flight_df["type"].value_counts()
```

```
type
Boeing                                    416
Cessna                                    366
Airbus                                    234
Antonov                                   184
Beechcraft                                172
                                         ...
North American Rockwell Sabreliner 60       2
de Havilland Canada DHC-8-202Q              2
Transall C-160NG                            2
Beriev Be-200ChS                            2
PZL-Mielec M28 Skytruck                     2
Name: count, Length: 76, dtype: int64
```

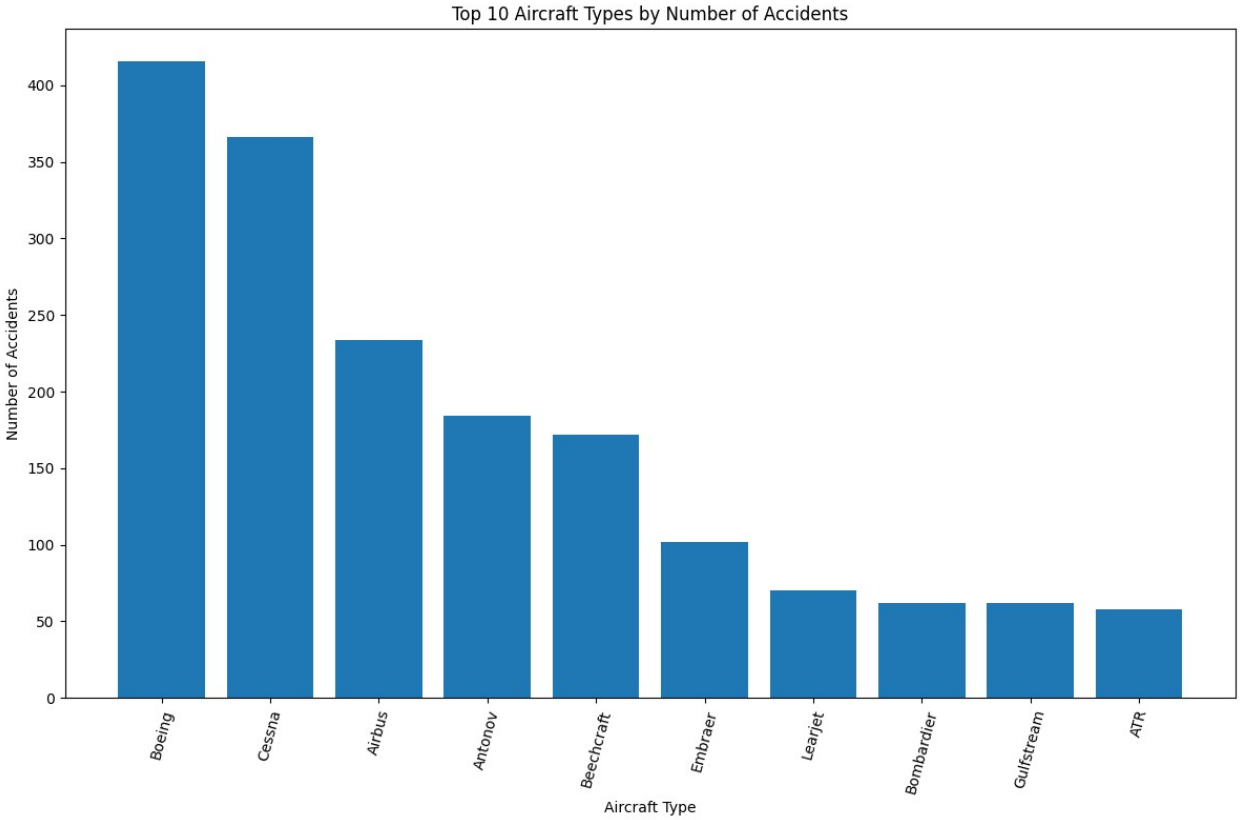
5. Data Analysis

Here we answer questions posed, that will help investors make decisions.

5.1 Descriptive Analysis

This helps us answer the question 'What happened?'. We ask questions like, how many accidents happened by aircraft type? and accidents over time (years).

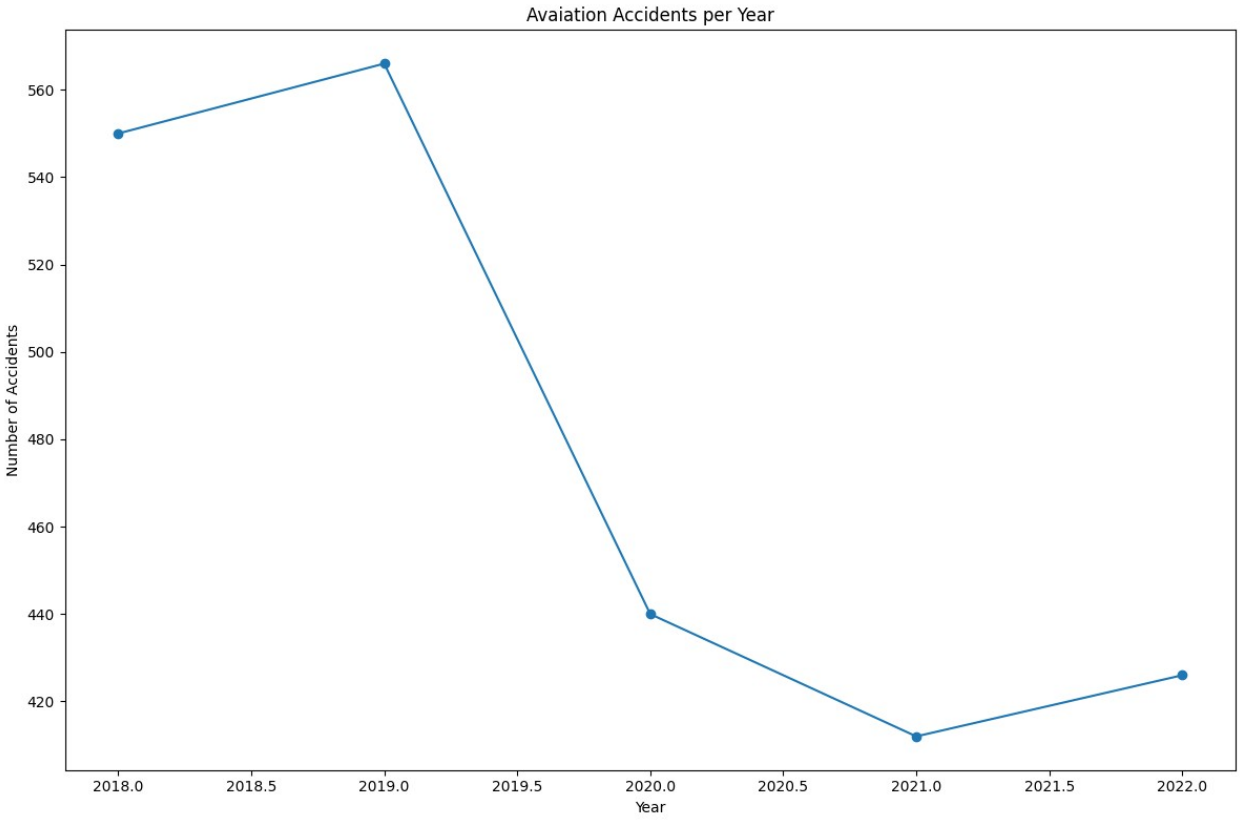
```
# Count accidents by aircraft type
type_accidents = flight_df['type'].value_counts()

# Select top 10 aircraft types with most accidents
top_10_types = type_accidents.head(10)

plt.figure()
plt.bar(top_10_types.index, top_10_types.values)
plt.xlabel('Aircraft Type')
plt.ylabel('Number of Accidents')
plt.title('Top 10 Aircraft Types by Number of Accidents')
plt.xticks(rotation=75)
plt.tight_layout()
plt.show()
```



The charts show that aircraft types with larger and more widely used fleets, such as Boeing and Cessna, appear more frequently in accident records. This likely reflects higher exposure and operational volume rather than inherently lower safety. That is why it would be best to analyze fatalities by aircraft type.

```
# Convert acc.date to datetime format
flight_df['acc.date'] = pd.to_datetime(
    flight_df['acc.date'],
    format='mixed',
    dayfirst=True,
    errors='coerce'  # Handle invalid dates by setting them to NaT
)
# Extract year from acc.date
flight_df['acc.year'] = flight_df['acc.date'].dt.year

# Count accidents per year
accidents_per_year = flight_df.groupby('acc.year').size()

# Plot line chart of accidents per year
plt.figure()
plt.plot(accidents_per_year.index, accidents_per_year.values,
marker='o')
plt.xlabel('Year')
plt.ylabel('Number of Accidents')
plt.title('Avaiation Accidents per Year')
```

```
plt.tight_layout()
plt.show()
```



Time series trend shows that:

1. 2018 -2019 (Slight increase):

This maybe be likey due to increase in air traffic and more flights that equals more exposure.

1. 2019 - 2020 (Sharp Drop):

Due to the COVID-19 pandemic, flights were grounded due to quarantine meaning fewer flights, fewer takeoffs and landings. Accidents drop due to reduced operations.

1. 2020 - 2021 (Continued decline):

Aviation still recovering with limited routes and fewer fights.

1. 2021 - 2022:

Air travel resumes, fleets returning to service and exposure increases again. Reflects the gradual recovery of aviation activity rather than a deterioration in safety.

5.2 Diagnostic Analysis

This helps us answer the question 'Why did it happen?'. We investigate relationship in the data ato explain patterns seen in descriptive analysis. Key areas of focus include comparing aircraft

types with the number of fatalities to determine whether certain aircraft are associated with more severe outcomes, as well as analyzing the relationship between damage severity and fatalities.

By examining these relationships, the analysis moves beyond simple accident counts and provides insight into factors that may influence accident severity.

```
# Confirm data type of 'fat' column
flight_df['fat'].dtype
```

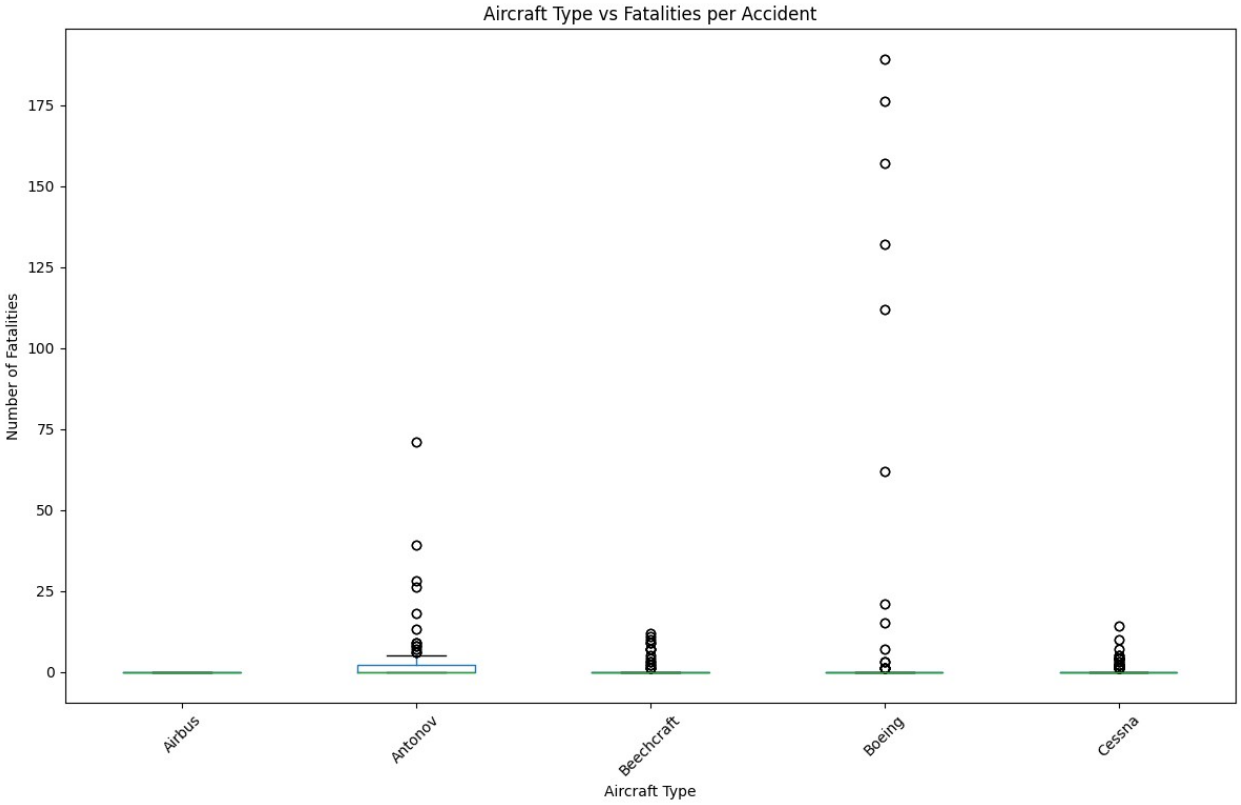
```
dtype('O')
```

```
flight_df['fat'] = pd.to_numeric(flight_df['fat'], errors='coerce')
flight_df['fat'].isna().sum()
```

```
np.int64(38)
```

```
# Get top 5 aircraft tpes by number of accidents
top_5_types = flight_df['type'].value_counts().head(5).index

# Filter DataFrame for top 5 aircraft types
top_df = flight_df[flight_df['type'].isin(top_5_types)]

# Plot accidents per year for top 5 aircraft types
plt.figure()
top_df.boxplot(
    column ='fat',
    by='type',
    grid=False
)

plt.xlabel("Aircraft Type")
plt.ylabel("Number of Fatalities")
plt.title("Aircraft Type vs Fatalities per Accident")
plt.suptitle("")  # Removes automatic pandas title
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

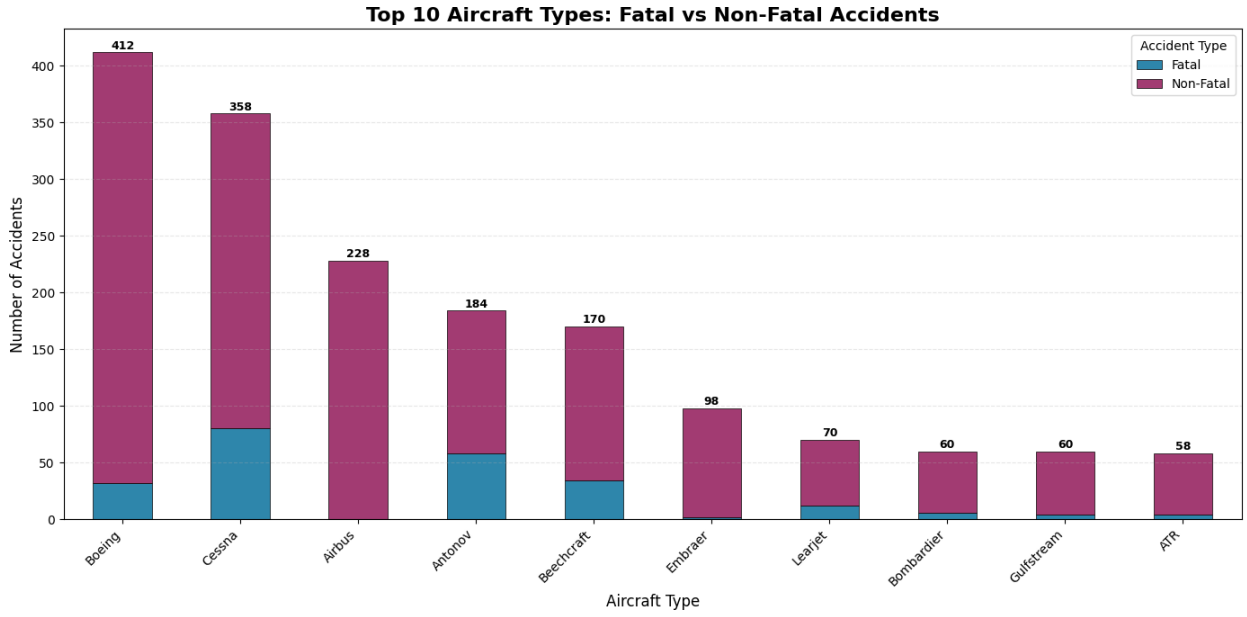
```
<Figure size 1200x800 with 0 Axes>
```



The box plot illustrates the distribution of fatalities per accident across major aircraft types. While the median number of fatalities is zero for all aircraft types, indicating that most accidents are non-fatal, some aircraft types such as Boeing and Antonov exhibit greater variability and more extreme outliers. This suggests that when accidents involving larger or specialized aircraft occur, they may result in higher fatality counts due to passenger capacity or operational conditions.

```
# Clean column names
flight_df.columns = flight_df.columns.str.strip()

# Convert fatalities to numeric
flight_df['fat'] = pd.to_numeric(flight_df['fat'], errors='coerce')

# Drop invalid rows
flight_df = flight_df.dropna(subset=['fat', 'type'])

# Create a new column for Fatal / Non-Fatal
flight_df['acc_type'] = flight_df['fat'].apply(lambda x: 'Fatal' if x
> 0 else 'Non-Fatal')

# Count accidents per type
stacked_data = flight_df.groupby(['type',
'acc_type']).size().unstack(fill_value=0)

# Get top 10 aircraft types by total accidents
```

```
stacked_data['Total'] = stacked_data.sum(axis=1)
top_10 = stacked_data.nlargest(10, 'Total').drop('Total', axis=1)

# Plot stacked bar chart
ax = top_10.plot(kind='bar', stacked=True, figsize=(14, 7),
                 color=['#2E86AB', '#A23B72'], edgecolor='black',
linewidth=0.5)
plt.title('Top 10 Aircraft Types: Fatal vs Non-Fatal Accidents',
fontsize=16, fontweight='bold')
plt.xlabel('Aircraft Type', fontsize=12)
plt.ylabel('Number of Accidents', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.legend(title='Accident Type', fontsize=10)
plt.grid(axis='y', alpha=0.3, linestyle='--')

# Add total count labels on top of bars
for i, (idx, row) in enumerate(top_10.iterrows()):
    total = row.sum()
    ax.text(i, total + 0.5, f'{int(total)}', ha='center', va='bottom',
fontweight='bold', fontsize=9)

plt.tight_layout()
plt.show()
```



The data reveals significant pattern in aviation safety across different aircraft categories. Boeing leads with total accidents, but remarkably a small precentage in fatalities, reflecting the robust safety systemsin modern commercial jets and the manufacturer's extensive operational fleet.

This pattern extends across the dataset: commercial manufacturers like Boeing, Airbus, and Embraer demonstrate lower fatality rates due to advanced safety features, rigorous training

requirements, and multiple backup systems, while general aviation manufacturers such as Cessna and Beechcraft show higher fatality proportions despite fewer overall accidents.

The data underscores that total accident counts don't necessarily correlate with danger, context matters significantly. Boeing's high accident count reflects decades of operations across thousands of aircraft, while the survivability rate demonstrates substantial progress in aviation safety engineering. Overall, the findings highlight how aircraft size, purpose, and safety systems dramatically influence accident outcomes in modern aviation.

```
# Save the cleaned DataFrame to a new CSV file
flight_df.to_csv('cleaned_flight_data.csv', index=False)
test_df = pd.read_csv('cleaned_flight_data.csv')
```

6. Conclusion

Prioritize Boeing and Airbus aircraft for fleet acquisition.

These manufacturers offer an ideal balance between operational safety and financial risk management. While their aircraft come with **higher upfront costs**, their long-term valu, driven by outstanding **survivability, robust safety systems, and proven reliability**, ensures superior return on investment while shielding stakeholders from catastrophic financial losses and reputational harm.